

Part 1

Problem 1

1. $(0) \subseteq \mathbb{Z}$
2. $(4) \subseteq \mathbb{Z}$
3. $\mathbb{R}[x_1, x_2, x_3, \dots]$
4. a
5. The map $\mathbb{Z} \rightarrow \mathbb{Z}$ given by $x \mapsto 0$